



A Project Report On “AI-Based Surgical Bite Size and Suturing Consistency Assessment Using YOLOv8”

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DECLARATION

We hereby declare that the matter embodied in this report entitled “AI-Based Surgical Bite Size and Suturing Consistency Assessment Using YOLOv8” submitted to KLE Technological University for the completion of Deep Learning (25ECAC305) in the 6th Semester of Computer Science and Engineering is the result of the work carried out by us in the Department of Computer Science and Engineering, KLE Technological University, Dr. M. S. Sheshgiri Campus, Belagavi, under the guidance of Dr. Salma Shahapur. We further declare that this work is original and has not been submitted, either in part or in full, for the award of any degree, diploma, or academic qualification in this or any other University or Institute.

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CERTIFICATE

This is to certify that the project entitled “**AI-Based Surgical Bite Size and Suture Consistency Assessment Using YOLOv8**” submitted to KLE Technological University’s Dr. MSSCET, Belagavi, for the partial fulfillment of the requirements of the course Deep Learning (25ECAC305) by Aditya Deshpande, Shrushti Hulbutte, Bhagyavathi Shetty, and Krishna Patel, students of the Department of Computer Science and Engineering, KLE Technological University, Dr. M. S. Sheshgiri Campus, Belagavi, is a bona fide record of the work carried out by them under my supervision.

The contents of this report, in full or in part, have not been submitted to any other Institute or University for the award of any degree, diploma, or course completion certificate.

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0.1 Abstract

Surgical suturing is one of the most fundamental skills in surgical practice, requiring precision, consistency, and proper tissue handling to ensure successful wound closure and patient recovery. Traditional assessment of suturing performance is primarily based on expert observation and subjective evaluation, which can be time-consuming and may vary between evaluators. The increasing adoption of artificial intelligence and computer vision technologies in healthcare has created new opportunities for developing objective and automated methods for surgical skill assessment.

This project presents an AI-Based Surgical Suturing Assessment System that utilizes deep learning and computer vision techniques to automatically detect surgical landmarks and perform quantitative analysis of suturing performance. Surgical procedure videos were collected from JNMC Medical College and Vijay Hospital and subsequently processed through a frame extraction pipeline to generate image datasets suitable for deep learning applications. The extracted frames were manually annotated to identify two clinically relevant surgical landmarks, namely the Sutured region and the Pull-Through region. The annotated data were organized according to the YOLO format and prepared for object detection model training.

The YOLOv8 object detection framework was employed to learn the spatial and visual characteristics of the annotated surgical landmarks. During training, the model optimized localization and classification performance through iterative learning and validation. The trained model demonstrated high detection accuracy and was capable of reliably identifying surgical landmarks across diverse surgical scenes. Standard evaluation metrics including Precision, Recall, F1-Score, and Mean Average Precision (mAP) were utilized to assess model performance and validate detection reliability.

In addition to object detection, the proposed framework incorporated a quantitative assessment module for estimating bite size using the geometric relationship between detected surgical landmarks. Bite-size measurements were calculated using the Euclidean distance between the centers of the detected Sutured and Pull-Through regions. Statistical analysis of these measurements enabled the evaluation of suturing consistency and provided objective insights into stitch placement characteristics. This approach transforms visual surgical information into measurable performance indicators that can be

used for skill assessment and training purposes.

The experimental results demonstrate that the proposed system successfully combines video processing, data annotation, deep learning-based object detection, and quantitative analysis into a unified framework for automated surgical assessment. By reducing dependence on subjective evaluation and providing objective performance metrics, the developed system has the potential to support surgical education, improve training outcomes, and contribute to the advancement of intelligent healthcare technologies. The proposed framework establishes a foundation for future research in AI-assisted surgical skill assessment and computer vision-based healthcare applications.

0.2 Introduction

Surgical suturing is one of the most essential technical skills in modern surgical practice. It plays a critical role in wound closure, tissue approximation, hemostasis, and postoperative healing. Proper suturing technique ensures adequate wound strength, minimizes tissue trauma, reduces the risk of infection, and promotes faster patient recovery. The quality of suturing directly influences surgical outcomes, making it a fundamental competency that every surgeon must master during training. Achieving proficiency in suturing requires extensive practice, continuous assessment, and expert feedback to ensure that surgical trainees develop the necessary precision and consistency required in clinical environments.

Traditionally, the assessment of surgical suturing skills relies heavily on direct observation by experienced surgeons and educators. Trainees are evaluated based on factors such as stitch placement, bite size, suture spacing, tissue handling, knot security, and overall procedural efficiency. Although expert evaluation remains the gold standard for surgical training, this approach is inherently subjective and may vary depending on the experience, judgment, and evaluation criteria of individual assessors. Furthermore, manual assessment is time-consuming and often difficult to perform consistently across large numbers of trainees. These limitations have created a growing demand for objective, quantitative, and automated methods for evaluating surgical performance.

Recent advances in artificial intelligence, machine learning, and computer vision have transformed numerous fields within healthcare. AI-based systems have demonstrated remarkable success in medical image analysis, disease diagnosis, surgical navigation, robotic surgery, and clinical decision support. Computer vision techniques enable machines to

interpret and analyze visual information from images and videos, allowing automated extraction of clinically relevant features that were previously assessed manually. The integration of AI into surgical education has created new opportunities for developing intelligent systems capable of objectively evaluating surgical skills and providing real-time feedback to trainees.

Among various computer vision techniques, deep learning-based object detection has emerged as one of the most effective approaches for identifying and localizing objects within images. Object detection models not only determine the presence of an object but also provide precise spatial information regarding its location. In recent years, the YOLO (You Only Look Once) family of object detection models has gained significant popularity due to its high accuracy, computational efficiency, and real-time processing capabilities. The latest version, YOLOv8, offers improved feature extraction, enhanced localization accuracy, and superior detection performance compared to earlier architectures. These characteristics make YOLOv8 particularly suitable for medical image analysis and surgical video assessment applications.

The availability of digital surgical recordings has further accelerated research in AI-assisted surgical assessment. Surgical videos contain valuable information regarding instrument movements, tissue interactions, stitch placement, and procedural techniques. By analyzing these videos using computer vision algorithms, it is possible to extract objective measurements that can be used to evaluate surgical performance. Automated assessment systems can reduce evaluator workload, improve consistency, and provide standardized feedback that supports effective skill development.

In this project, surgical procedure videos were collected from JNMC Medical College and Vijay Hospital to develop an automated framework for surgical suturing assessment. The collected videos were processed through a frame extraction pipeline to generate individual image frames suitable for computer vision analysis. These frames were manually annotated to identify clinically relevant suturing landmarks. Two important classes were considered during annotation: the Sutured region and the Pull-Through region. These landmarks represent key components of the suturing process and serve as the basis for subsequent quantitative analysis.

Following annotation, the prepared dataset was used to train a YOLOv8-based object detection model capable of automatically identifying the selected surgical landmarks.

The trained model learns the visual characteristics of the annotated regions and generates bounding boxes around the detected landmarks during inference. By accurately localizing these landmarks, the system can extract geometric information required for further assessment of suturing performance.

Beyond object detection, the proposed framework incorporates quantitative analysis techniques to evaluate suturing quality. One of the primary assessment parameters considered in this study is bite size estimation. Bite size represents the distance between detected surgical landmarks and serves as an indicator of stitch placement characteristics. Consistent bite sizes generally reflect controlled tissue handling and uniform suturing technique, whereas large variations may indicate inconsistencies in performance. By analyzing bite-size measurements across multiple frames, the system can provide objective insights into suturing consistency and technical proficiency.

The overall workflow of the proposed system consists of several stages, including video acquisition, frame extraction, data annotation, dataset preparation, YOLOv8 model training, object detection, bite-size estimation, and suturing consistency analysis. Each stage contributes to the development of a comprehensive framework for automated surgical skill evaluation. The integration of deep learning and computer vision enables the transformation of visual surgical information into measurable performance indicators that can support both trainees and instructors.

The proposed AI-Based Surgical Suturing Assessment System aims to address the limitations of traditional subjective evaluation methods by providing an objective, reliable, and scalable assessment solution. By automating the detection of surgical landmarks and generating quantitative performance metrics, the system contributes to the advancement of intelligent surgical training technologies. Furthermore, the developed framework demonstrates the potential of artificial intelligence in enhancing healthcare education and creating data-driven approaches for surgical skill assessment.

Overall, this project highlights the application of modern deep learning techniques in the field of surgical education and healthcare analytics. The successful implementation of automated suturing assessment demonstrates how artificial intelligence can assist in improving training quality, standardizing evaluation procedures, and supporting the development of highly skilled surgical professionals.

0.3 Literature Survey

The current research findings show that deep learning, computer vision, and artificial intelligence techniques can be effectively used for automated surgical skill assessment and suturing analysis. The use of convolutional neural networks, object detection models, and temporal learning frameworks has significantly improved the accuracy and efficiency of surgical evaluation systems. However, challenges such as dataset limitations, real-time implementation, scalability, robustness, and generalization still exist. It can thus be concluded from the literature that intelligent surgical assessment systems are rapidly emerging in the healthcare domain.

Mansour et al., [1] proposed a deep learning-based framework for evaluating suture images used in surgical training and achieved nearly 95% accuracy using CNN models. Tao et al., [2] developed Sparse Hidden Markov Models for surgical skill classification and demonstrated high performance on gesture datasets. Hu et al., [3] introduced real-time surgical skill assessment using temporal machine learning and deep learning models on the JIGSAWS dataset. Zia et al., [4] presented an automated assessment framework for robotic minimally invasive surgery using machine learning techniques for accurate classification.

Wang et al., [5] implemented CNN-based surgical skill assessment models and achieved approximately 95% accuracy with online evaluation capability. Khalid et al., [6] proposed an encoder–decoder deep learning framework for surgical skill assessment and suggested the use of larger datasets for improved performance. Liu et al., [7] developed a multi-path deep learning framework for surgical video analysis and improved evaluation metrics and generalization. Baby et al., [8] introduced a rank-based skill evaluation system using deep learning and rank loss methods for improved ranking performance.

Dhanakshirur et al., [9] proposed a micro-suturing skill assessment system using MT-CNN models for effective scoring and handling data imbalance issues. Handelman et al., [10] focused on morphological skill evaluation using image processing methods on the JIGSAWS dataset. Kil et al., [11] designed an image-based suturing skill assessment framework using computer vision techniques for real-time analysis. Sirajudeen et al., [12] developed deep learning models for robotic surgery skill and error prediction, achieving approximately 80% accuracy and an F1-score of nearly 59%.

Khairnar et al., [13] implemented machine learning models such as Random Forest, SVM, and CNN for laparoscopic skill assessment with nearly 81.7% accuracy. Hu et al., [14] proposed a multimodal deep learning framework for surgical skill assessment with fine-grained feedback mechanisms. Wang et al., [15] introduced time-series CNN models for surgical skill evaluation and suggested transfer learning for further enhancement. Dhanakshirur et al., [16] also developed automated micro-suturing evaluation frameworks using CNN models for effective scoring.

Kil et al., [17] proposed computer vision-based skill metrics capable of differentiating surgical skill levels using simulator datasets. Sirajudeen et al., [18] implemented MS-TCN models for surgical error detection and improved prediction capability. Khairnar et al., [19] developed YOLOv8-based tool tracking methods for surgical skill assessment using approximately 1400 images. Finally, Hu et al., [20] introduced a window-based real-time assessment framework using machine learning and deep learning models for continuous surgical feedback and performance evaluation.

0.4 Methodology

The proposed AI-Based Surgical Suturing Assessment System was developed to automatically detect surgical suturing landmarks and perform quantitative assessment of suturing quality using deep learning and computer vision techniques. The complete methodology consists of data acquisition, preprocessing, annotation, model training, evaluation, and suturing metric analysis. The workflow adopted in this project is illustrated through the following stages.

0.4.1 Data Collection

The dataset used in this study was collected from real surgical suturing procedures performed at **Jawaharlal Nehru Medical College (JNMC), Belagavi** and **Vijay Hospital, Belagavi**. The videos were recorded during surgical procedures under the supervision of medical professionals, ensuring that the captured data represented practical clinical scenarios. The collection process focused on obtaining clear visual recordings of suturing activities while maintaining patient confidentiality and adhering to ethical considerations.

The surgical videos were captured using high-resolution smartphone cameras positioned to provide a clear view of the wound closure region and suturing process. To maintain consistency across recordings, the camera zoom level was kept fixed at **2× optical zoom (52 mm focal length equivalent)** throughout the recording session. Keeping the zoom constant ensured uniform visual scale and minimized variations caused by camera movement or focal length changes.

The collected videos contained various stages of the suturing procedure, including needle insertion, tissue approximation, pull-through actions, and completed sutures. These videos captured different surgical environments, lighting conditions, viewing angles, and operator techniques, thereby providing a diverse dataset for model training and evaluation.

After data acquisition, the recorded videos were transferred to the processing environment for further analysis. Since consecutive video frames often contain highly similar visual information, representative frames were extracted at predefined intervals to reduce redundancy while preserving relevant surgical content. For one video sequence, every fourth frame was extracted, whereas for the remaining video sequences, every fifth frame was extracted. This approach significantly reduced the dataset size while retaining sufficient information required for object detection and suturing assessment.

The extracted frames were then manually reviewed and annotated to identify two clinically relevant suturing landmarks:

- **Sutured Region (Class 0)** – Represents the completed stitch area on the tissue surface.
- **Pull-Through Region (Class 1)** – Represents the point where the suture thread is being pulled through the tissue.

A total of five surgical video sequences were utilized for dataset preparation, generating thousands of extracted frames. Following annotation and dataset cleaning, only frames containing valid landmark annotations were retained for model training and evaluation. This ensured that the final dataset consisted of high-quality, clinically relevant samples suitable for developing an automated surgical suturing assessment system.

The collected dataset forms the foundation of the proposed framework and enables the development of a robust deep learning model capable of detecting key suturing landmarks

and supporting quantitative surgical skill assessment.

0.4.2 Frame Extraction

The surgical procedures were recorded as continuous video sequences collected from JNMC Medical College and Vijay Hospital. Since the YOLOv8 object detection model requires image-based inputs, the recorded videos were converted into individual image frames through a frame extraction process.

Frame extraction was performed using Python and OpenCV libraries. Extracting every frame from the video would result in a large number of redundant images because adjacent frames often contain nearly identical visual information. To reduce dataset redundancy while preserving meaningful surgical events, a fixed frame sampling strategy was adopted.

For the video sequence A001_04241410_C008, every fourth frame was extracted, resulting in image sequences such as:

00000, 00004, 00008, 00012, 00016, ...

For the remaining video sequences, namely B001_04141321_C013new, B001_04300947_C036, B001_05021237_C040, and B001_05021312_C041, every fifth frame was extracted, resulting in image sequences such as:

00005, 00010, 00015, 00020, 00025, ...

This sampling strategy significantly reduced computational complexity and storage requirements while retaining sufficient information required for surgical landmark detection and analysis. The extracted frames captured different stages of the suturing procedure, including needle insertion, tissue approximation, pull-through actions, and completed sutures.

Each frame was saved in JPEG format and assigned a unique frame identifier corresponding to its position within the original video sequence. The extracted images were organized into separate directories for each surgical video, ensuring proper traceability throughout the annotation and training pipeline.



(a) Extracted Frame 1



(b) Extracted Frame 2



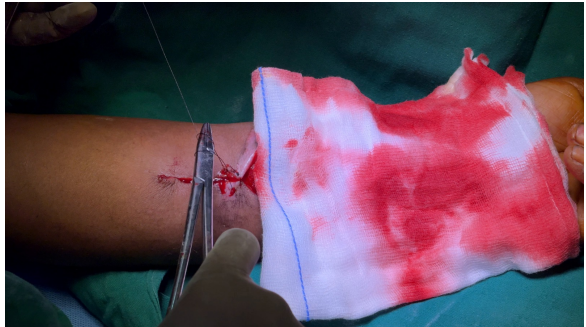
(c) Extracted Frame 3



(d) Extracted Frame 4



(e) Extracted Frame 4



(f) Extracted Frame 4

Figure 1: Representative surgical frames extracted from the recorded videos at fixed sampling intervals. The extracted frames capture different stages of the suturing procedure and provide the image data used for annotation and YOLOv8 model training.

0.4.3 Data Annotation

After frame extraction, the generated surgical images were manually annotated to create a labeled dataset suitable for training the YOLOv8 object detection model. Data annotation is a crucial step in supervised deep learning because it provides the ground truth information required for the model to learn meaningful visual features and accurately identify target objects.

The annotation process was performed by carefully examining each extracted surgical frame and identifying clinically relevant suturing landmarks. Two object classes were defined for the study:

- **Class 0: Sutured** – Represents the completed sutured region on the tissue.
- **Class 1: Pull-Through** – Represents the region where the suture thread is pulled through the tissue during the suturing process.

Each frame was manually annotated using a bounding box surrounding the target landmark. The annotation information was stored in YOLO format, where every object is represented by a single row containing five values:

Class_ID X_Center Y_Center Width Height

where:

- **Class_ID** represents the object category.
- **X_Center** represents the normalized horizontal center coordinate of the bounding box.
- **Y_Center** represents the normalized vertical center coordinate of the bounding box.
- **Width** represents the normalized width of the bounding box.
- **Height** represents the normalized height of the bounding box.

All coordinates were normalized with respect to image dimensions, ensuring compatibility with the YOLOv8 training framework. Corresponding annotation files were stored as text files with the same filename as the associated image.

To improve annotation quality, the generated labels were visually verified by comparing bounding boxes against the original surgical images. Frames containing incorrect or empty annotations were excluded during the dataset cleaning stage. This ensured that only valid and clinically meaningful samples were used for model training.

The final annotated dataset consisted of surgical images containing one or more labeled suturing landmarks. These annotations served as the ground truth data for training the YOLOv8 model to automatically detect sutured and pull-through regions in unseen surgical frames.

0.4.4 Dataset Cleaning and Organization

To improve training quality, only frames containing valid annotations were retained. Empty annotation files and their corresponding images were removed from the training dataset.

A structured dataset hierarchy was created:

```
clean_dataset/  
  A001_04241410_C008/  
    images/  
    labels/  
  B001_04141321_C013new/  
    images/  
    labels/  
  B001_04300947_C036/  
    images/  
    labels/  
  B001_05021237_C040/  
    images/  
    labels/  
  B001_05021312_C041/  
    images/  
    labels/
```

This organization ensured proper mapping between images and annotation files while preventing filename conflicts between different video sequences.

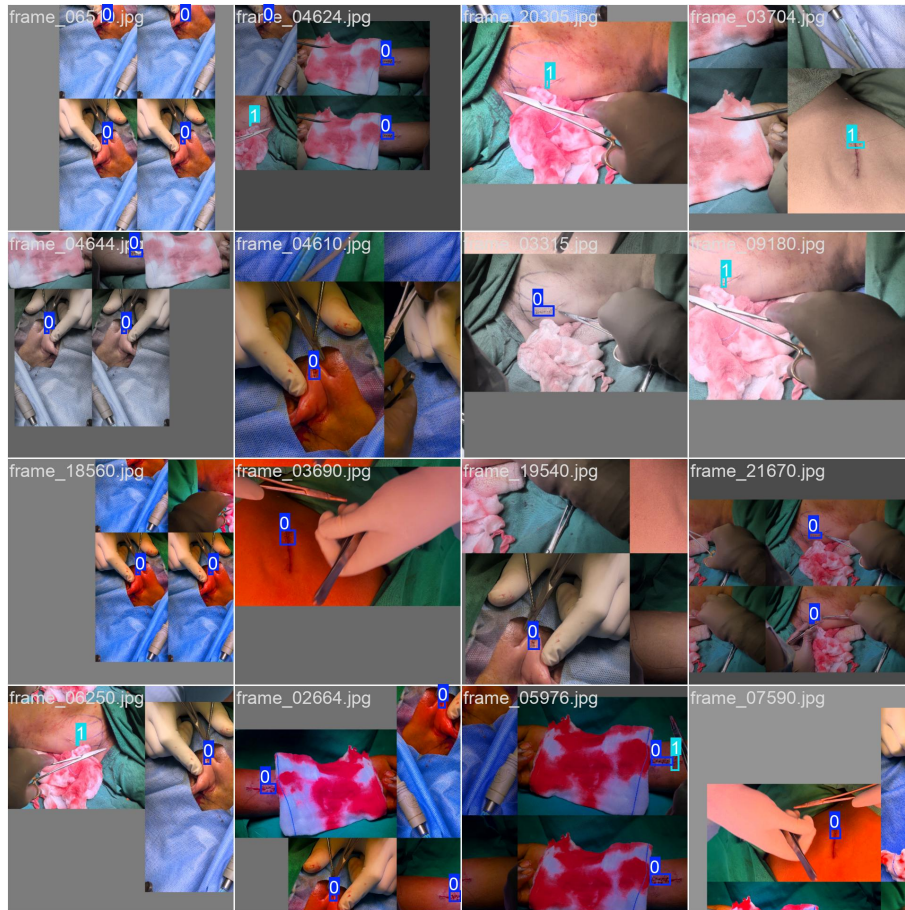


Figure 2: Annotated Frame 1

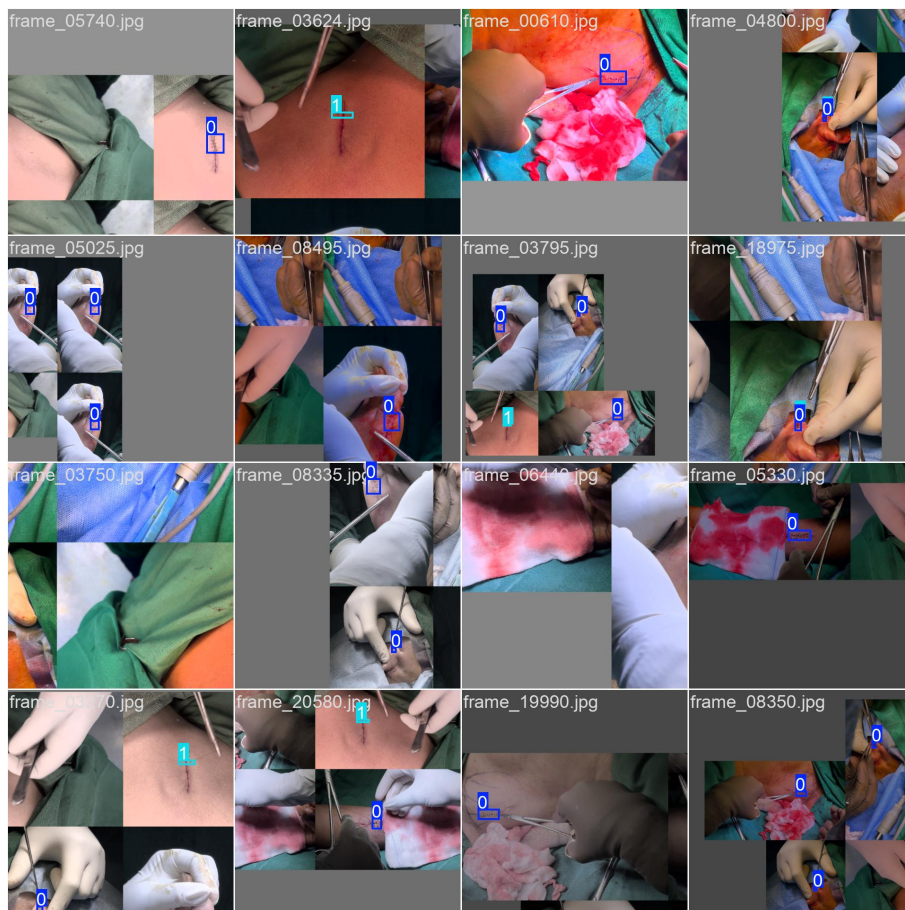


Figure 3: Annotated Frame 2



Figure 4: Annotated Frame 3

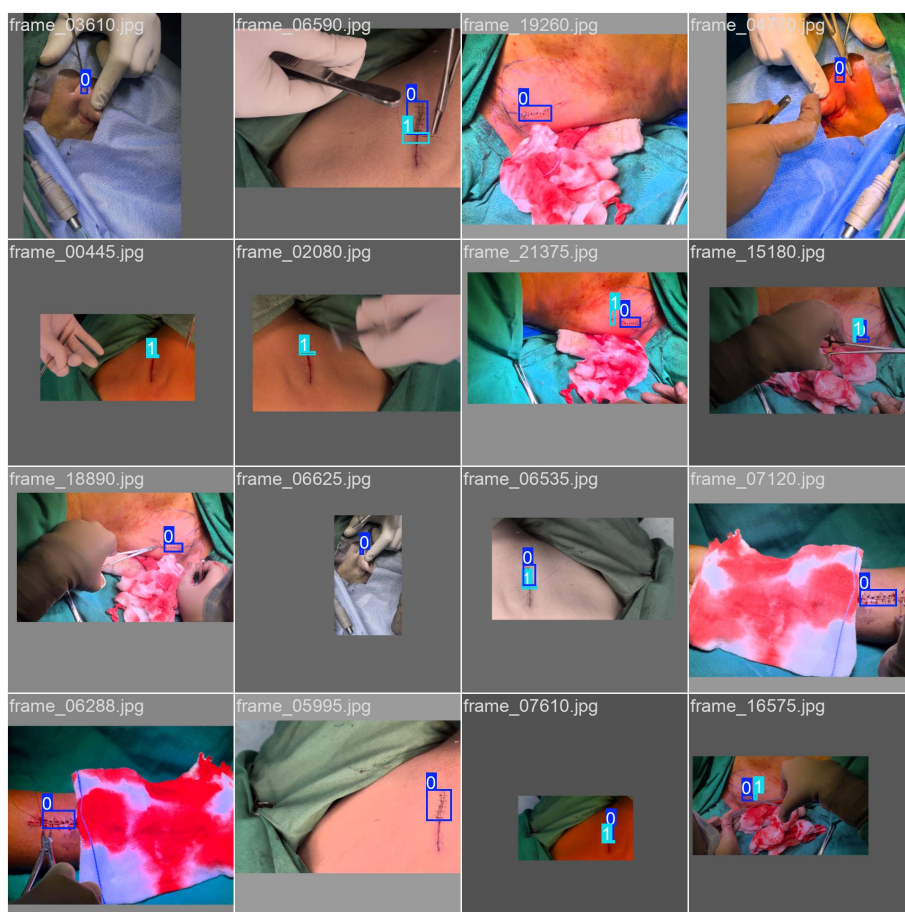


Figure 5: Annotated Frame 4

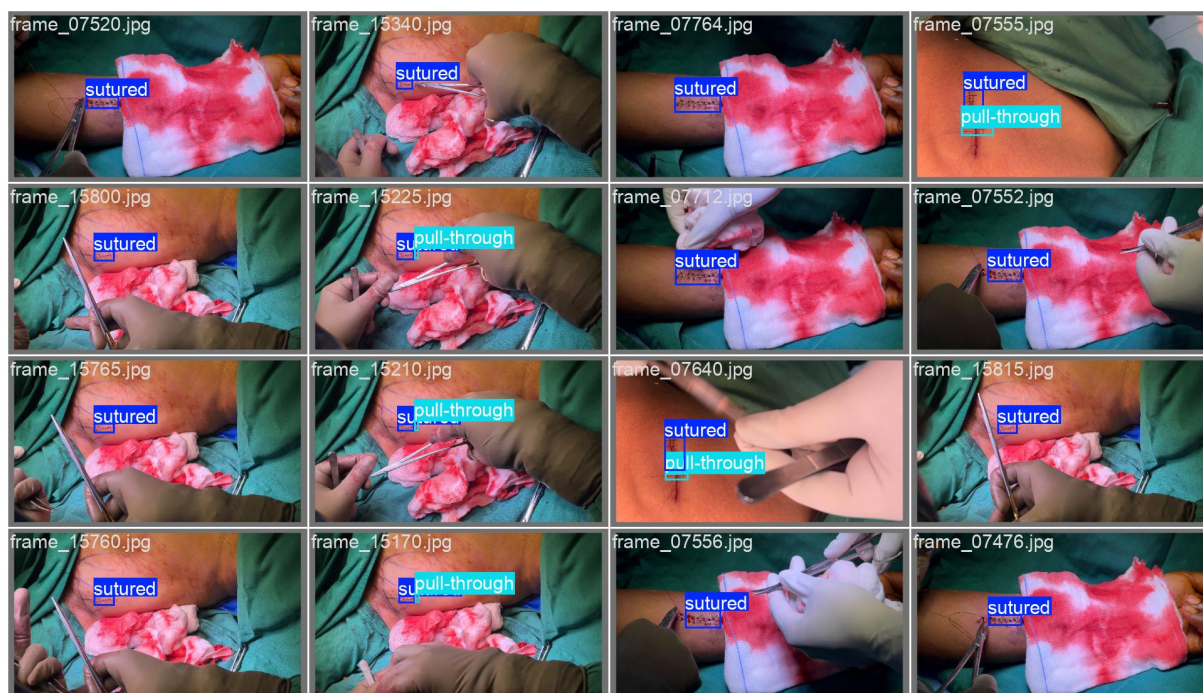


Figure 6: Annotated Frame 5

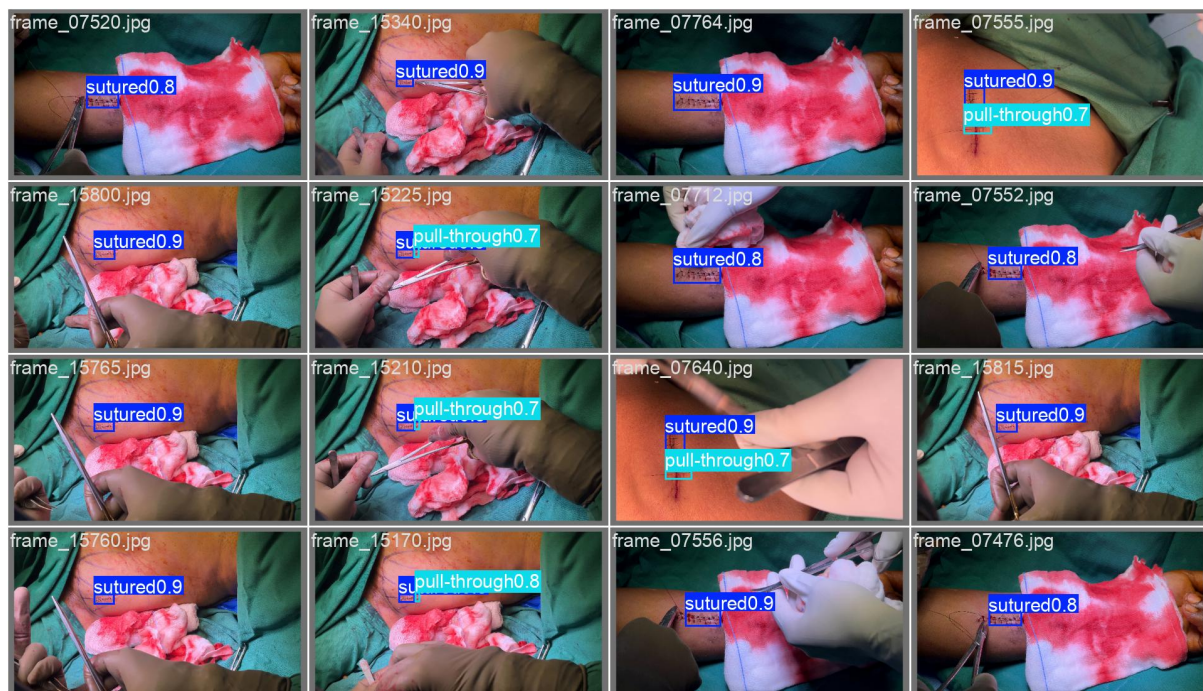


Figure 7: Annotated Frame 6

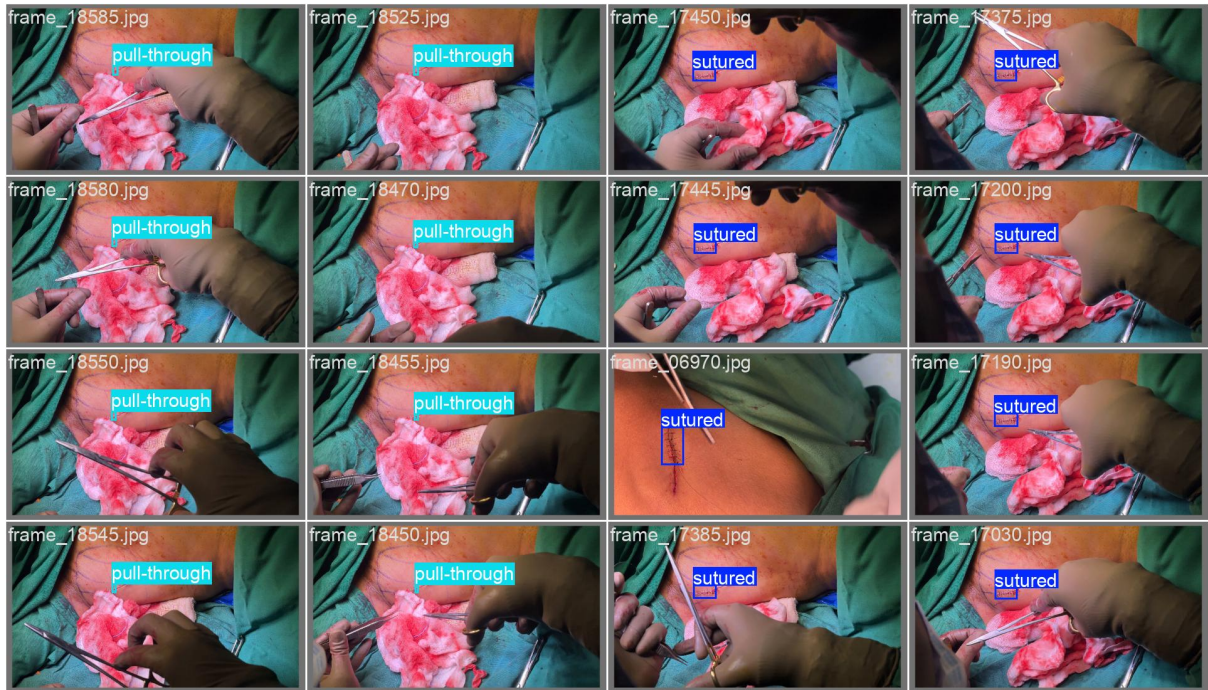


Figure 8: Annotated Frame 7

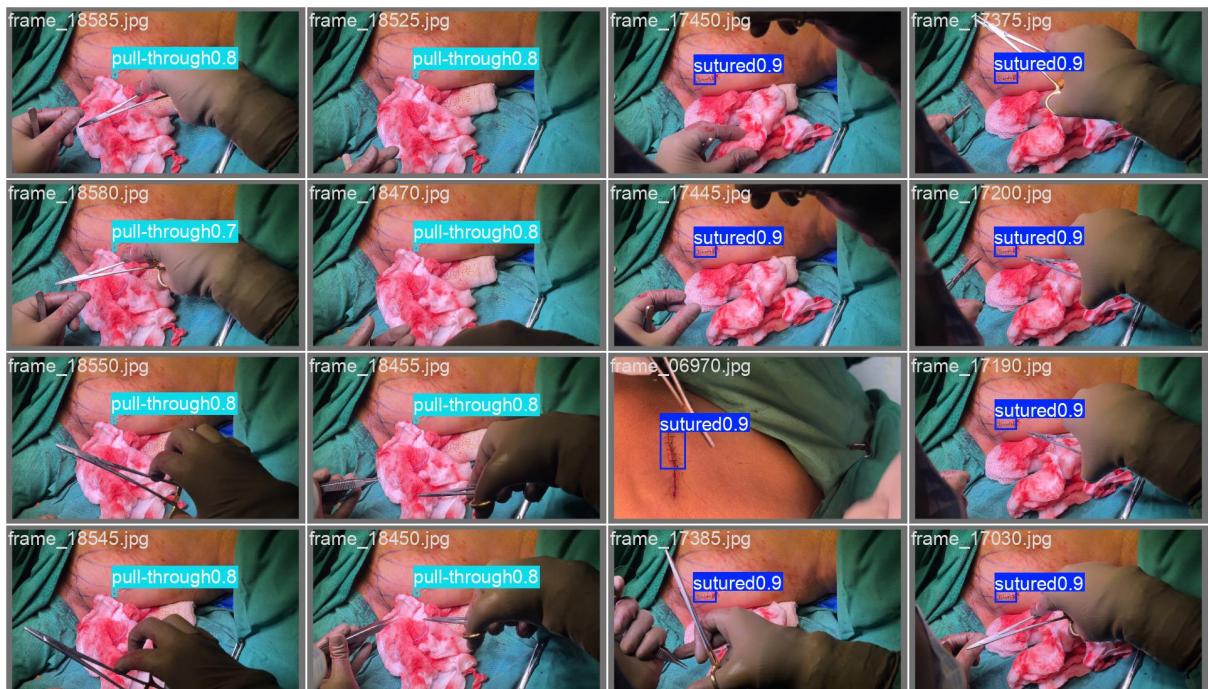


Figure 9: Annotated Frame 8

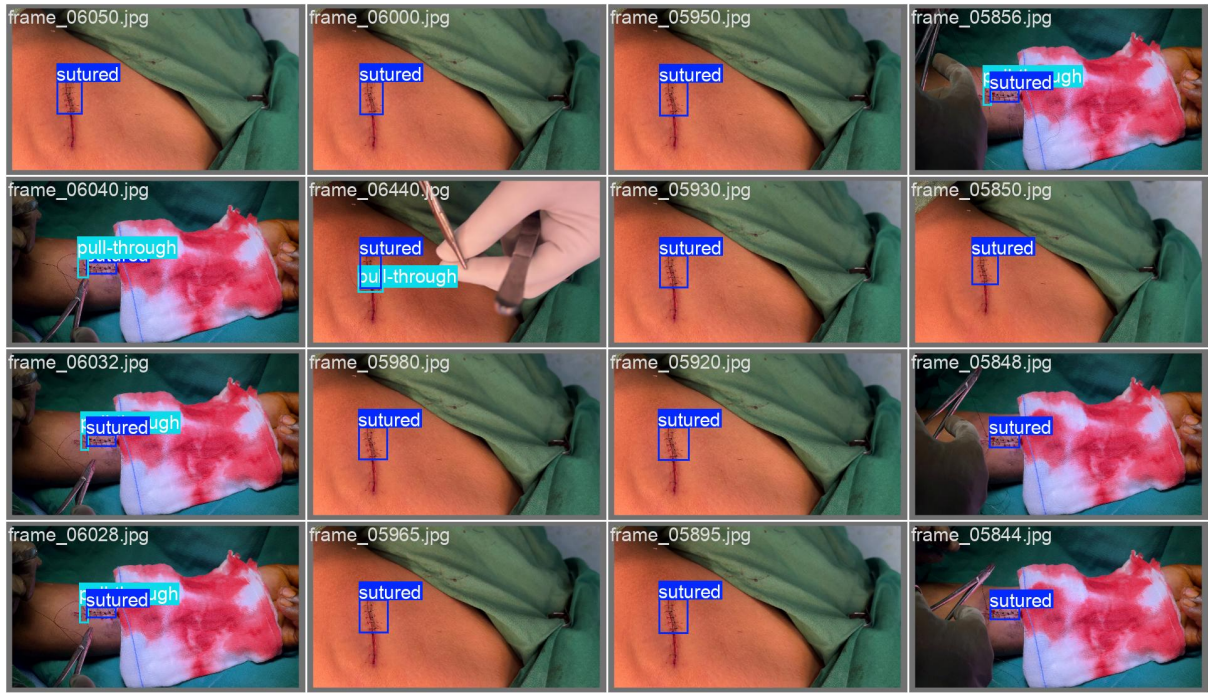


Figure 10: Annotated Frame 9

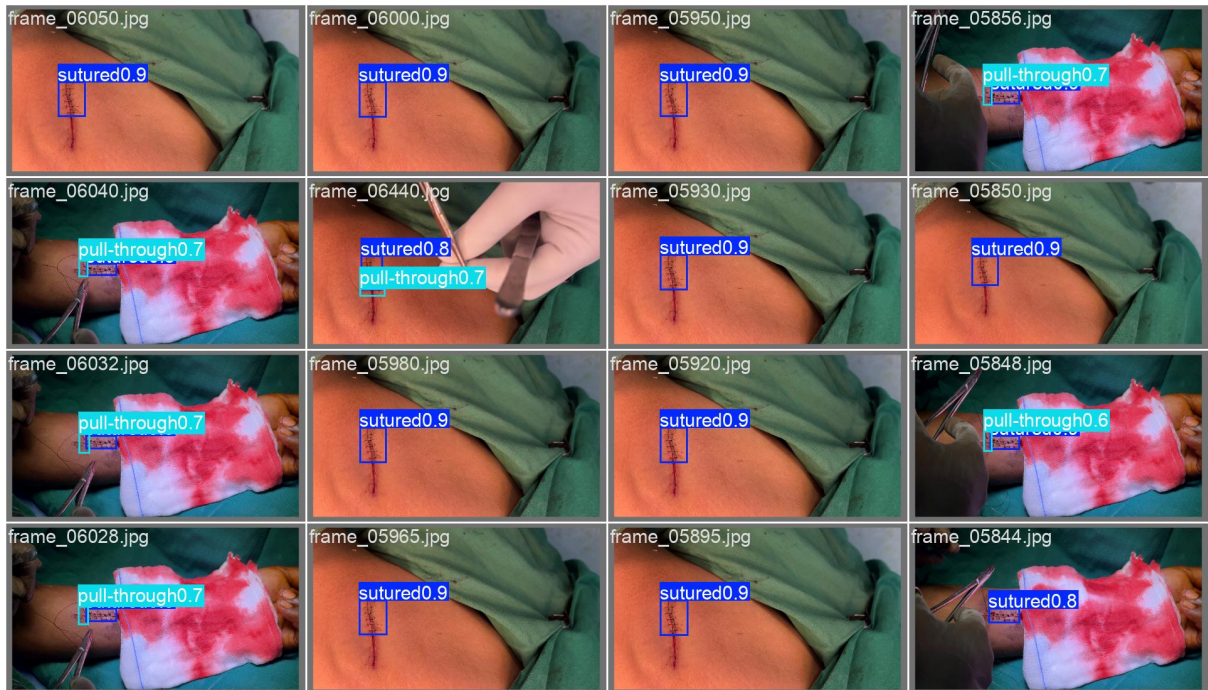


Figure 11: Annotated Frame 10

0.4.5 Dataset Preparation for YOLO

Following frame extraction, annotation, and dataset cleaning, the prepared surgical image dataset was organized according to the directory structure required by the YOLOv8 object detection framework. Proper dataset preparation is essential to ensure that the model can correctly associate images with their corresponding annotations during training and validation.

Initially, the extracted surgical frames and annotation files were organized separately for each surgical video sequence. Since multiple videos contained image files with similar naming conventions, maintaining a structured folder hierarchy was necessary to avoid filename conflicts and preserve traceability throughout the training pipeline.

The cleaned dataset was organized into the following structure:

```
clean_dataset/
```

```
A001_04241410_C008/
```

```
images/
```

```
labels/
```

```
B001_04141321_C013new/
```

```
images/
```

```
labels/
```

```
B001_04300947_C036/
```

```
images/
```

```
labels/
```

```
B001_05021237_C040/
```

```
images/
```

```
labels/
```

```
B001_05021312_C041/
```

```
images/
```

`labels/`

Each image stored within the `images` directory had a corresponding annotation file with the same filename located in the `labels` directory. This one-to-one mapping between images and annotation files ensured compatibility with the YOLOv8 training framework.

During dataset cleaning, all frames without valid annotations were removed. Empty label files and their corresponding images were excluded from the training dataset to prevent the model from learning irrelevant background information. This step improved dataset quality and reduced unnecessary computational overhead during training.

After cleaning, all valid images and annotation files were consolidated into the standard YOLO dataset structure consisting of training and validation subsets:

`dataset/`

`train/`

`images/`

`labels/`

`valid/`

`images/`

`labels/`

The dataset was then split into training and validation sets to evaluate model generalization on previously unseen data. The training set was used for learning the visual characteristics of the surgical landmarks, while the validation set was used to monitor model performance during training and detect potential overfitting.

A dataset configuration file (`data.yaml`) was created to provide YOLOv8 with the required dataset information. The configuration file contained the dataset paths, number of classes, and class names used in the project.

The `data.yaml` file was defined as follows:

`train: /dataset/train/images`

`val: /dataset/valid/images`

`nc: 2`

`names:`

`0: Sutured`

`1: Pull_Through`

`where:`

- **train** specifies the path to the training images.
- **val** specifies the path to the validation images.
- **nc** represents the total number of object classes.
- **names** defines the class labels used during training.

Once the dataset structure and configuration file were prepared, the data became fully compatible with the Ultralytics YOLOv8 framework. The prepared dataset was then utilized for model training, validation, and performance evaluation. This structured preparation process ensured efficient data loading, accurate image-label mapping, and reliable learning of surgical landmark features during model training.

0.4.6 Data Preparation for YOLO

Following frame extraction, annotation, and dataset cleaning, the prepared surgical image dataset was organized according to the directory structure required by the YOLOv8 object detection framework. Proper dataset preparation is essential to ensure that the model can correctly associate images with their corresponding annotations during training and validation.

Initially, the extracted surgical frames and annotation files were organized separately for each surgical video sequence. Since multiple videos contained image files with similar naming conventions, maintaining a structured folder hierarchy was necessary to avoid filename conflicts and preserve traceability throughout the training pipeline.

The cleaned dataset was organized into the following structure:

`clean_dataset/`

A001_04241410_C008/

images/

labels/

B001_04141321_C013new/

images/

labels/

B001_04300947_C036/

images/

labels/

B001_05021237_C040/

images/

labels/

B001_05021312_C041/

images/

labels/

Each image stored within the **images** directory had a corresponding annotation file with the same filename located in the **labels** directory. This one-to-one mapping between images and annotation files ensured compatibility with the YOLOv8 training framework.

During dataset cleaning, all frames without valid annotations were removed. Empty label files and their corresponding images were excluded from the training dataset to prevent the model from learning irrelevant background information. This step improved dataset quality and reduced unnecessary computational overhead during training.

After cleaning, all valid images and annotation files were consolidated into the standard YOLO dataset structure consisting of training and validation subsets:

dataset/

train/

```
images/  
labels/  
  
valid/  
  images/  
  labels/
```

The dataset was then split into training and validation sets to evaluate model generalization on previously unseen data. The training set was used for learning the visual characteristics of the surgical landmarks, while the validation set was used to monitor model performance during training and detect potential overfitting.

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```
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val: /dataset/valid/images  
  
nc: 2  
  
names:  
  0: Sutured  
  1: Pull_Through
```

where:

- **train** specifies the path to the training images.
- **val** specifies the path to the validation images.
- **nc** represents the total number of object classes.
- **names** defines the class labels used during training.

Once the dataset structure and configuration file were prepared, the data became fully compatible with the Ultralytics YOLOv8 framework. The prepared dataset was

then utilized for model training, validation, and performance evaluation. This structured preparation process ensured efficient data loading, accurate image-label mapping, and reliable learning of surgical landmark features during model training.

0.4.7 Model Validation and Evaluation

Model performance was evaluated using standard object detection metrics.

Precision

Precision measures the proportion of correctly detected objects among all predicted objects.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

Recall measures the proportion of correctly detected objects among all actual objects.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 Score

The F1 score provides a balanced measure of precision and recall.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Mean Average Precision (mAP)

Mean Average Precision (mAP) is one of the most widely used evaluation metrics for object detection models. It provides a comprehensive measure of a model's ability to accurately detect and localize objects within an image. Unlike simple classification metrics, mAP evaluates both the correctness of object classification and the accuracy of bounding box localization, making it particularly suitable for assessing object detection performance.

In object detection, a prediction is considered correct only when the predicted class matches the ground-truth class and the predicted bounding box sufficiently overlaps with the corresponding ground-truth bounding box. The degree of overlap between the predicted and actual bounding boxes is measured using the Intersection over Union (IoU) metric.

The IoU is calculated as:

$$[\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}]$$

where the area of overlap represents the common region shared by the predicted and ground-truth bounding boxes, while the area of union represents the total combined area covered by both bounding boxes.

A prediction is classified as a True Positive when the IoU exceeds a predefined threshold and the object class is correctly identified. Predictions with insufficient overlap or incorrect class assignments are treated as False Positives, while undetected objects are counted as False Negatives.

To evaluate detection performance, Precision and Recall are calculated as follows:

$$[\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}]$$

$$[\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}]$$

where:

- TP = True Positives
- FP = False Positives
- FN = False Negatives

By varying the detection confidence threshold, multiple Precision and Recall values are obtained. These values are used to generate a Precision–Recall curve, which illustrates the trade-off between precision and recall across different confidence levels.

The Average Precision (AP) for a particular class is computed as the area under the Precision–Recall curve. A higher AP value indicates better detection performance for that specific class.

When multiple object classes are present, the Average Precision values of all classes are averaged to obtain the Mean Average Precision (mAP):

$$[\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i]$$

where:

- (N) represents the total number of classes.
- (AP_i) represents the Average Precision of the (i^{th}) class.

In YOLOv8, two commonly reported mAP metrics are:

- **mAP@0.5:** Evaluates detection performance at an IoU threshold of 0.5.
- **mAP@0.5:0.95:** Computes the average mAP across multiple IoU thresholds ranging from 0.5 to 0.95 with an interval of 0.05.

The mAP@0.5 metric measures general detection capability, while mAP@0.5:0.95 provides a stricter evaluation by considering increasingly demanding localization requirements. Consequently, mAP@0.5:0.95 is often regarded as a more comprehensive measure of object detection performance.

In this project, mAP was used as one of the primary evaluation metrics to assess the effectiveness of the YOLOv8 model in detecting the Sutured and Pull-Through surgical landmarks. The metric provided quantitative insight into the model's ability to accurately localize and classify surgical regions, thereby ensuring reliable performance for automated suturing assessment.

0.4.8 Bite Size Estimation

Following successful detection of the surgical landmarks, a quantitative analysis stage was performed to estimate bite size from the detected Sutured and Pull-Through regions. Bite size is an important suturing parameter that describes the relative distance between successive tissue penetration points and provides insight into the consistency and quality of the suturing technique. Uniform bite sizes generally indicate controlled tissue handling and consistent stitch placement, whereas large variations may suggest irregular suturing performance.

In the proposed system, bite size estimation was performed using the spatial coordinates obtained from the YOLOv8 object detection model. For each detected object, YOLOv8 generates the center coordinates of the corresponding bounding box. Since the dataset contained two annotated landmark classes, namely Sutured and Pull-Through regions, the center coordinates of these detected landmarks were extracted for further analysis.

Let:

$[(x_1, y_1)]$

represent the center coordinates of the detected Sutured region and

$[(x_2, y_2)]$

represent the center coordinates of the detected Pull-Through region.

The bite size was estimated by calculating the Euclidean distance between these two landmark centers. The Euclidean distance provides a measure of the straight-line separation between the detected surgical landmarks and is computed as:

$$[\text{Bite Size} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}]$$

where:

- $(x_1) = \text{Horizontal coordinate of the Sutured landmark}$
- $(y_1) = \text{Vertical coordinate of the Sutured landmark}$
- $(x_2) = \text{Horizontal coordinate of the Pull - Through landmark}$
- $(y_2) = \text{Vertical coordinate of the Pull - Through landmark}$

The resulting distance value represents the relative bite size for a particular surgical stitch within the analyzed frame. Since YOLO annotations utilize normalized coordinates, the calculated bite-size values are also expressed in normalized image units rather than absolute physical measurements.

The bite-size estimation procedure was applied to all valid frames containing both Sutured and Pull-Through detections. Frames in which one or both landmarks were absent were excluded from the calculation process to ensure reliable measurements. For each valid frame, a single bite-size value was computed and stored for subsequent statistical analysis.

After obtaining bite-size measurements from all analyzed frames, descriptive statistical parameters were calculated, including:

- Mean Bite Size
- Standard Deviation
- Minimum Bite Size

- Maximum Bite Size

The mean bite size provides an estimate of the average stitch spacing observed during the procedure, while the standard deviation measures the variability of bite-size measurements. Lower standard deviation values indicate greater consistency in stitch placement, whereas higher values suggest increased variation in suturing technique.

To further evaluate suturing performance, the calculated bite-size measurements were analyzed across individual surgical videos. This video-wise analysis enabled comparison of stitch consistency between different procedures and provided objective quantitative information regarding surgical execution.

The bite-size estimation framework developed in this project transforms the detected landmark coordinates into meaningful surgical assessment metrics. By replacing subjective visual inspection with measurable geometric analysis, the proposed approach provides a reliable and repeatable method for evaluating suturing quality. The extracted bite-size measurements form the foundation for subsequent consistency assessment and contribute to the overall objective evaluation of surgical performance.

0.4.9 Suturing Consistency Analysis

Following bite-size estimation, a suturing consistency analysis was performed to evaluate the uniformity of stitch placement throughout the surgical procedure. Consistency is an important indicator of surgical skill because experienced surgeons generally maintain similar bite sizes across successive stitches, resulting in uniform wound closure and improved tissue approximation. In contrast, significant variations in stitch placement may indicate inconsistencies in technique and reduced procedural precision.

The proposed system assessed suturing consistency by analyzing the bite-size measurements obtained from all valid surgical frames. Each bite-size value represented the Euclidean distance between the detected Sutured and Pull-Through landmarks within a particular frame. These measurements were collected and statistically analyzed to determine the degree of variation present during the suturing process.

The first step of the analysis involved calculating the mean bite size, which represents the average distance observed across all analyzed stitches. The mean bite size was computed using:

$$[\mu = \frac{1}{N} \sum_{i=1}^N B_i]$$

where:

- (N) represents the total number of valid bite-size measurements.
- (B_i) represents the bite size of the (i^{th}) stitch.

The mean value provides a reference point for evaluating the overall stitch spacing maintained throughout the procedure.

To quantify the variability of bite sizes, the standard deviation was calculated as:

$$[\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (B_i - \mu)^2}]$$

where:

- (σ) represents the standard deviation.
- (μ) represents the mean bite size.

The standard deviation measures how closely individual bite-size values are distributed around the mean. A lower standard deviation indicates that the stitches are more uniformly spaced, whereas a higher standard deviation suggests greater variation in suturing technique.

In addition to the mean and standard deviation, the minimum and maximum bite-size values were computed to determine the range of stitch variation observed during the procedure. These values provide additional insight into the smallest and largest stitch distances produced by the surgeon.

To provide an easily interpretable measure of suturing quality, a consistency score was calculated based on the relationship between the mean bite size and its variation. The consistency score reflects the degree of uniformity in stitch placement throughout the analyzed surgical sequence. Higher scores indicate greater consistency and more controlled suturing behavior, while lower scores indicate increased variability and less uniform stitch placement.

The consistency analysis was performed both globally and on a video-wise basis. Evaluating individual surgical videos separately allowed comparison of suturing behavior across different procedures and helped identify variations in performance. Video-wise statistical analysis provided detailed information regarding the consistency of stitch placement within each surgical recording.

The generated consistency metrics provide objective quantitative feedback that can assist in surgical skill evaluation. Unlike traditional assessment methods that rely primarily on expert observation, the proposed approach utilizes measurable geometric characteristics extracted from detected surgical landmarks. This enables repeatable and unbiased analysis of suturing performance.

Overall, suturing consistency analysis serves as an important component of the proposed assessment framework. By evaluating the uniformity of bite-size measurements across multiple stitches, the system provides meaningful insight into surgical precision, tissue handling control, and overall suturing quality.

0.5 Results and Discussion

0.5.1 Training Performance Analysis

The YOLOv8 model was trained for 50 epochs using the annotated surgical image dataset. During training, the model learned to detect two surgical landmarks, namely the Sutured region and Pull-Through region. The training process demonstrated stable convergence with continuously decreasing loss values and improving evaluation metrics.

The box loss, classification loss, and distribution focal loss decreased steadily throughout training, indicating successful learning of localization and classification features. Simultaneously, precision, recall, mAP@0.5, and mAP@0.5:0.95 showed consistent improvement, confirming effective model optimization.

The final model achieved approximately 98% precision, 97% recall, 97% F1-score, and 98.3% mAP@0.5, demonstrating excellent detection performance for surgical landmark identification.

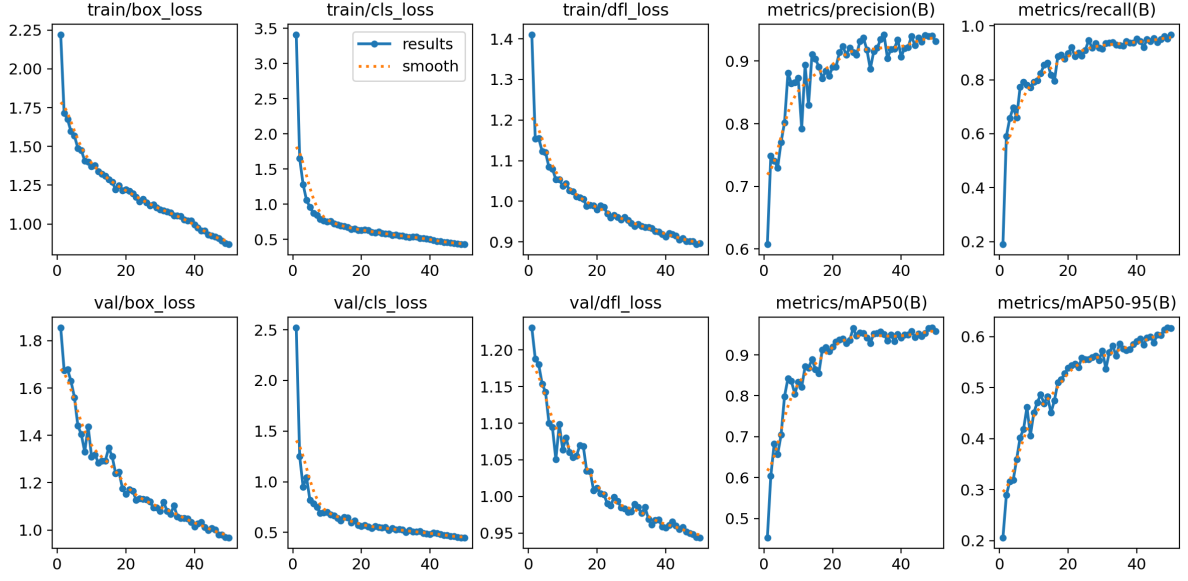


Figure 12: Training and validation performance of the YOLOv8 model over 50 epochs showing reduction in loss values and improvement in precision, recall, mAP@0.5, and mAP@0.5:0.95 metrics.

0.5.2 Precision-Recall Analysis

The Precision-Recall (PR) curve evaluates the detection capability of the trained model across varying confidence thresholds. The curve demonstrates strong detection performance for both surgical landmark classes.

The Sutured class achieved an Average Precision (AP) of approximately 99.5%, while the Pull-Through class achieved an AP of approximately 97.2%. The overall mean Average Precision at an IoU threshold of 0.5 was approximately 98.3%.

The PR curves remain close to the upper-right corner of the graph, indicating excellent precision and recall simultaneously. These results confirm the reliability and robustness of the proposed model for surgical landmark detection.

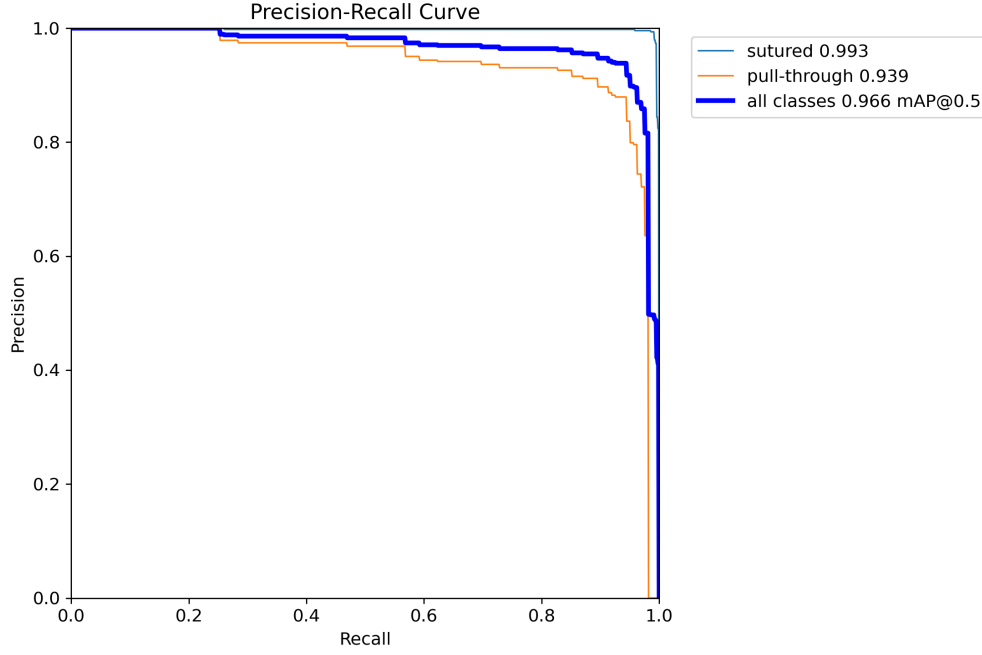


Figure 13: Precision-Recall curve showing detection performance of the Sutured and Pull-Through classes.

0.5.3 F1-Confidence Analysis

The F1-Confidence curve illustrates the relationship between confidence thresholds and overall detection performance. The maximum F1 score achieved by the model was approximately 0.97 at a confidence threshold of approximately 0.43.

This result indicates that the selected confidence threshold provides an optimal balance between precision and recall. The high F1 score further validates the effectiveness of the trained YOLOv8 model in identifying surgical landmarks.

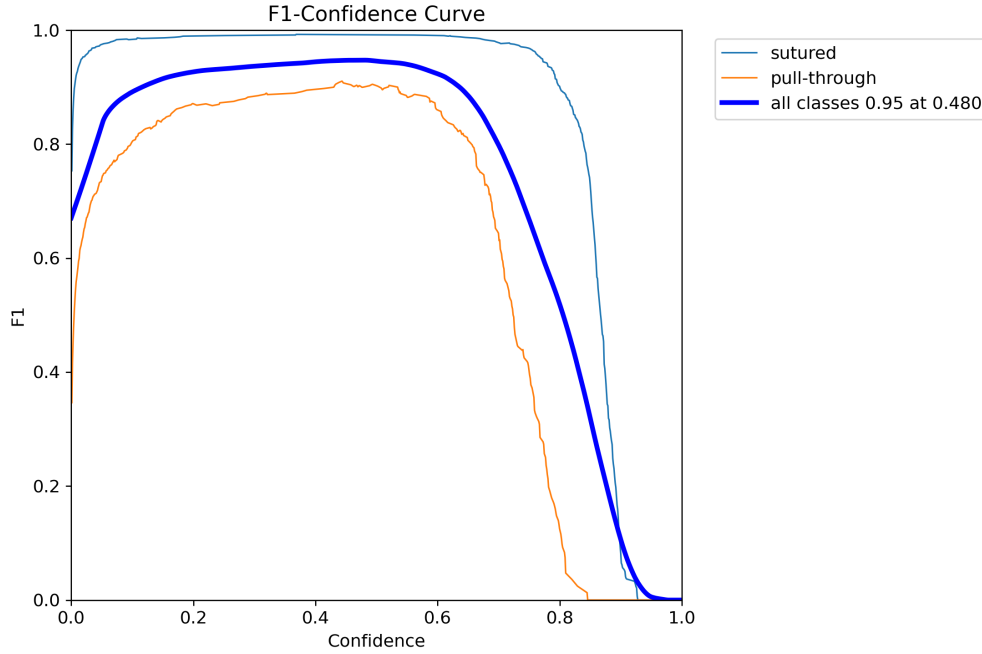
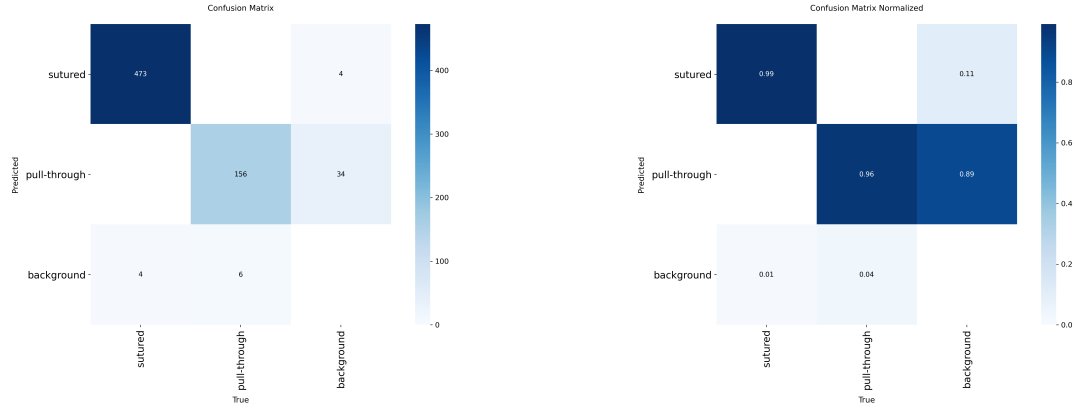


Figure 14: F1-Confidence curve showing the optimal confidence threshold for maximum detection performance.

0.5.4 Confusion Matrix Analysis

The confusion matrix provides a detailed evaluation of class-wise prediction performance. The trained model successfully classified the majority of Sutured and Pull-Through instances with very few misclassifications.

Analysis of the confusion matrix indicates that the model accurately distinguishes between the two surgical landmark classes while maintaining low false-positive and false-negative rates. Only a small number of detections were incorrectly assigned to the background class, demonstrating strong generalization capability and reliable performance across diverse surgical scenes.



(a) Confusion Matrix

(b) Normalized Confusion Matrix

Figure 15: Confusion matrix analysis showing class-wise prediction performance of the YOLOv8 model.

0.5.5 Object Detection Results

After completing the training process, the performance of the YOLOv8 model was evaluated on the validation dataset to determine its effectiveness in detecting surgical landmarks. The primary objective of the object detection stage was to accurately identify and localize the two predefined classes, namely the Sutured region and the Pull-Through region, within surgical image frames.

The trained model successfully detected the target landmarks across a wide variety of surgical scenes, demonstrating strong generalization capability despite variations in tissue appearance, illumination conditions, viewing angles, and procedural stages. For each input image, the model generated bounding boxes surrounding the detected landmarks along with confidence scores indicating the probability of correct detection.

The object detection process involved two fundamental tasks: object localization and object classification. Localization refers to identifying the precise position of a surgical landmark within the image using a bounding box, while classification refers to assigning the correct class label to the detected region. The YOLOv8 model performed both tasks simultaneously, enabling efficient and accurate surgical landmark detection.

The evaluation results demonstrated that the model achieved high detection accuracy, indicating that the learned features effectively represented the visual characteristics of the Sutured and Pull-Through regions. During inference, the model was able to identify the target landmarks even in the presence of complex backgrounds, surgical instruments,

tissue texture variations, and partial occlusions.

Several quantitative performance metrics were used to evaluate the detection capability of the model. These included Precision, Recall, F1-Score, and Mean Average Precision (mAP). Precision measures the proportion of correctly detected objects among all predicted detections, while Recall measures the proportion of actual objects that were successfully detected. The F1-Score provides a balanced evaluation of Precision and Recall, whereas Mean Average Precision evaluates both localization accuracy and classification performance.

The trained YOLOv8 model achieved the following performance:

- Precision: 98
- Recall: 97
- F1-Score: 97
- mAP@0.5: 98.3

The high Precision value indicates that the majority of detected landmarks corresponded to actual surgical regions, resulting in very few false-positive detections. Similarly, the high Recall value demonstrates that most ground-truth landmarks present in the images were successfully detected by the model.

The F1-Score of 97

The Precision-Recall analysis further validated the effectiveness of the detector. The generated Precision-Recall curves remained close to the ideal upper-right region, indicating that the model maintained high precision while simultaneously achieving high recall across different confidence thresholds. Similarly, the F1-Confidence analysis identified an optimal confidence threshold that maximized detection performance.

The confusion matrix provided additional insight into class-wise detection behavior. Results showed that the majority of Sutured and Pull-Through instances were correctly classified, with only a small number of misclassifications. The low number of false positives and false negatives demonstrates the robustness of the trained detector and its ability to accurately differentiate between the two surgical landmark classes.

Visual inspection of detection outputs further confirmed the effectiveness of the proposed approach. The generated bounding boxes closely aligned with the annotated surgical landmarks, indicating accurate localization performance. The model consistently

identified both classes across different surgical frames, demonstrating reliable detection even under challenging imaging conditions.

The successful detection of Sutured and Pull-Through regions forms the foundation for subsequent quantitative analysis. The coordinates extracted from the detected bounding boxes were utilized for bite-size estimation and suturing consistency assessment. Therefore, the object detection stage serves as the core component of the proposed surgical assessment framework, enabling automated extraction of clinically relevant information from surgical images.

Overall, the obtained object detection results demonstrate that the YOLOv8 model is highly effective for surgical landmark identification and provides a reliable basis for automated surgical skill assessment. The high detection accuracy achieved in this study highlights the potential of deep learning-based object detection techniques for intelligent healthcare and surgical training applications.

0.5.6 Bite Size Analysis

Following object detection, the coordinates of the detected Sutured and Pull-Through regions were extracted to estimate bite size. The Euclidean distance between the centers of the detected landmarks was calculated for each valid frame.

A total of 491 valid bite-size measurements were obtained from the analyzed surgical frames. The statistical analysis of bite size measurements produced the following results:

- Mean Bite Size = 0.0579
- Standard Deviation = 0.0347
- Minimum Bite Size = 0.0140
- Maximum Bite Size = 0.1477

The bite-size distribution revealed noticeable variation across different surgical procedures, reflecting differences in suturing technique and tissue handling characteristics.

Video-wise bite-size statistics are summarized below:

Table 1: Video-wise Bite Size Statistics

Video	Mean	Std	Min	Max	Count
A001_04241410_C008	0.0566	0.0152	0.0364	0.0917	107
B001_04141321_C013new	0.1041	0.0273	0.0663	0.1477	121
B001_04300947_C036	0.0622	0.0161	0.0354	0.0886	20
B001_05021237_C040	0.0216	0.0035	0.0140	0.0291	130
B001_05021312_C041	0.0508	0.0171	0.0213	0.0929	113

The analysis shows that different surgical procedures exhibited varying bite-size distributions. Lower variability generally indicates greater consistency in stitch placement, whereas larger variations suggest less uniform suturing behavior.

0.5.7 Suturing Consistency Assessment

To evaluate suturing consistency, statistical variation in bite-size measurements was analyzed across all detected stitches. A consistency score was calculated based on the variability of bite-size measurements.

The obtained consistency score was:

$$\text{Consistency Score} = 40.14 / 100$$

This score indicates moderate variability in stitch placement and bite-size selection throughout the analyzed procedures. The consistency analysis demonstrates the feasibility of using automatically detected landmarks to generate objective performance metrics for surgical skill assessment.

0.5.8 Discussion

The experimental results demonstrate that deep learning-based object detection can effectively identify clinically relevant suturing landmarks in surgical images. The YOLOv8 model achieved high precision, recall, F1-score, and mAP values, confirming its suitability for automated surgical skill assessment applications.

The generated bite-size measurements provide an objective alternative to conventional subjective evaluation methods. Instead of relying solely on expert observation,

the proposed framework enables quantitative assessment based on measurable geometric characteristics extracted directly from surgical images.

The high detection accuracy achieved by the YOLOv8 model ensures reliable localization of sutured and pull-through regions, which serves as the foundation for subsequent performance analysis. Furthermore, the bite-size estimation framework demonstrates the potential for extracting clinically meaningful metrics from automatically detected landmarks.

The obtained results highlight the feasibility of integrating artificial intelligence and computer vision techniques into surgical training environments. By providing automated detection, quantitative analysis, and objective feedback, the proposed system can assist trainees in improving technical proficiency while reducing the workload associated with manual evaluation.

Overall, the proposed AI-Based Surgical Suturing Assessment System successfully combines deep learning, object detection, and quantitative analysis to provide an efficient, reliable, and scalable framework for intelligent surgical skill assessment.

0.6 Conclusion

This project presented an AI-Based Surgical Suturing Assessment System that utilizes deep learning and computer vision techniques to automate the evaluation of surgical suturing performance. The primary objective of the study was to develop an efficient and objective framework capable of identifying clinically relevant suturing landmarks and extracting quantitative metrics that can assist in surgical skill assessment.

The proposed system was developed using surgical video recordings collected from JNMC Medical College and Vijay Hospital. The recorded videos were converted into image frames through a systematic frame extraction process, after which the extracted frames were manually annotated to identify two important surgical landmarks: the Sutured region and the Pull-Through region. The annotated dataset was then cleaned, organized, and prepared for training using the YOLOv8 object detection framework.

The YOLOv8 model successfully learned the visual characteristics of the defined surgical landmarks and demonstrated excellent detection performance during evaluation. Experimental results showed a Precision of approximately 98%, Recall of 97%, F1-Score of 97%, and mAP@0.5 of 98.3%, indicating reliable and accurate landmark localization

across diverse surgical scenes. The confusion matrix, Precision-Recall curves, and F1-Confidence analysis further confirmed the robustness and generalization capability of the trained model.

Beyond object detection, the detected landmark coordinates were utilized to perform quantitative suturing analysis. Bite-size measurements were computed using the Euclidean distance between the detected Sutured and Pull-Through regions. Statistical analysis of these measurements enabled the assessment of suturing consistency and stitch placement characteristics. A total of 491 valid bite-size measurements were analyzed, providing objective insights into surgical performance that would traditionally require manual expert evaluation.

The obtained results demonstrate that artificial intelligence can significantly contribute to surgical training by providing automated, objective, and reproducible assessment methods. The developed framework reduces dependence on subjective observations and enables the extraction of measurable performance indicators from surgical images. Such capabilities can support trainees in improving their technical skills while assisting instructors in monitoring surgical proficiency more efficiently.

Overall, the proposed AI-Based Surgical Suturing Assessment System successfully integrates video processing, data annotation, deep learning-based object detection, and quantitative analysis into a unified framework for surgical skill evaluation. The study establishes the feasibility of applying computer vision techniques to automate surgical assessment and provides a strong foundation for future intelligent healthcare and surgical training applications.

0.7 Future Scope

Although the proposed system successfully detects surgical landmarks and performs bite-size analysis, several opportunities exist for further enhancement. Future work may focus on estimating real-world inter-suture distances using camera calibration techniques and pixel-to-millimeter conversion methods. Additional surgical assessment parameters such as stitch spacing, bite symmetry, needle entry angle, knot positioning, and wound edge approximation can also be incorporated to provide a more comprehensive evaluation of suturing quality.

Furthermore, advanced deep learning architectures and segmentation-based approaches

may be explored to improve landmark localization accuracy. The framework can also be extended for real-time assessment during live surgical procedures and integrated into intelligent surgical training platforms. Deployment on edge devices and mobile healthcare systems would further enhance accessibility and practical usability in clinical and educational environments. These advancements have the potential to transform traditional surgical training by enabling continuous, objective, and AI-assisted skill assessment.

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